

# Deepanjay Nandal

Bengaluru, India | +91 8199074272 | [deepanjay.nandal12@gmail.com](mailto:deepanjay.nandal12@gmail.com) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## SUMMARY

Senior Software Engineer with 3+ years building systems, from CVE validation infrastructure for Google's Android Security team to LLM inference caching and event-driven pipelines processing 100k+ daily events. Combines low-level systems and security depth with AI infrastructure across Kubernetes, semantic caching, hybrid retrieval, and Kafka/Redis pipelines.

## EXPERIENCE

### Discover Excellence LLC

California, USA

*Software Engineer (Contract)*

Jan 2026 – Jun 2026

- Built a high-throughput Kafka-based visitor intelligence pipeline for an AI-powered eCommerce analytics platform, processing 100k+ daily behavioral events at sub-40ms p95 latency to drive discount outreach via AWS SES.
- Served personalized product feeds at the edge via Cloudflare, backed by Redis collaborative filtering, achieving 92% cache hit rate and sub-30ms p95 latency for multi-tenant deployments.
- Implemented a fault-tolerant BullMQ background pipeline for async job processing, handling 50k+ daily jobs with guaranteed job delivery through persistent state, retries, exponential backoff, and idempotency keys.

### Ittiam Systems Pvt Ltd

Bengaluru, India

*Senior Software Engineer*

Mar 2022 – Jun 2024

- Authored 50+ CTS/STS validation tests merged into the Android Open Source Project (AOSP), each proving a CVE patch for Google's Android Security team, working directly with Google engineers.
- Developed a Ghidra-based binary analysis approach for the Android Bluetooth stack that Google had marked infeasible, which the team adopted to validate 40+ Bluetooth CVEs.
- Engineered a Frida instrumentation framework in Python and Java to hook private Android methods at runtime, enabling the team to reproduce 30+ CVEs that standard testing could not reach.
- Built a GCP Python pipeline ingesting ClusterFuzz crash outputs across Dockerized environments, auto-classifying reports by CVSS severity into a PostgreSQL-backed schema, cutting manual triage overhead by 35%.
- Automated webhook-driven Gerrit CI/CD workflows with Change-Id-keyed idempotency, processing 30–40 CVE submissions per month without duplicate change creation.

## PROJECTS

### BitMod (AI Inference Cache) | *Python, FastAPI, Vector Embeddings, Multi-tenant, Docker*

Jan – Jun 2026

- Architected a drop-in 9-layer LLM cache proxy with SHA-256 exact lookup, semantic reuse, and Bayesian evidence scoring, achieving 76% hit rate on a 50-query benchmark (100% on repeats, 53% on rephrasings).
- Held zero false serves across 375 held-out queries with nothing cached, reached 90–101x speedup over self-hosted LLM calls at sub-90ms cached, and enforced multi-tenant namespace isolation in SQL.
- Designed cost-aware LRU eviction scoring entries by predicted future hits against original LLM spend.

### CaseTally (Legal Search Platform) | *FastAPI, pgvector, RAG, Kubernetes, KEDA*

Jun – Dec 2025

- Constructed a RAG over 83k+ U.S. Code chunks with LLM issue decomposition, fusing PostgreSQL full-text and pgvector HNSW via RRF at MRR 0.85 and P@3 0.78 while cutting p50 latency from 325ms to 45ms (7x).
- Migrated the stack to Kubernetes with a Postgres StatefulSet, Traefik ingress, probes and advisory-locked ingestion Job, achieving zero-downtime rollouts (240 requests, 0 failures) and a 73% smaller backend image.
- Built an embedding queue on FOR UPDATE SKIP LOCKED with KEDA scaling workers 0 to 3 on queue depth, verifying 3,000/3,000 rows with zero loss or duplication under SIGKILL and raising worker throughput 55%.

### WeVibe | *Backend Engineer, Appy.yo (Capstone) | Node.js, PostgreSQL, Socket.IO, Redis*

Jan – Apr 2026

- Created a real-time matchmaking backend using Socket.IO and Redis pub/sub for horizontally scalable event delivery, PostgreSQL advisory locks preventing race conditions, and PostGIS geo-filtering on Google Cloud Run.
- Integrated Gemini bio generation with a custom prompt-injection guard and rate limiting (5 bios/day, 60s cooldown), Firebase JWT auth, and Apple Sign-In with revocation, validated across 17 Jest integration suites.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, Java, SQL

**Backend Technologies:** Distributed Systems, Microservices, Event-Driven Architecture, Node.js, Multi-tenancy, Concurrency Control, REST APIs, WebSockets, Message Queues, BullMQ, Rate Limiting, Socket.IO

**Data & Storage:** Apache Kafka, PostgreSQL, PostGIS, SQLite, Redis, pgvector, Qdrant, HNSW, Distributed Caching

**Security & Systems:** Android Open Source Project (AOSP), Binary Analysis, Ghidra, Frida, ClusterFuzz

**Cloud & DevOps:** Kubernetes, KEDA, Docker, CI/CD, AWS (EC2, Lambda, SQS, IoT Greengrass), GCP, Cloudflare

**AI Infrastructure:** Retrieval-Augmented Generation (RAG), Vector Search, Reciprocal Rank Fusion (RRF), Large Language Model (LLM) APIs, Semantic Caching, Embeddings, Collaborative Filtering, Retrieval Evaluation

## EDUCATION

### Arizona State University

Tempe, Arizona, USA

*Master of Science (M.S.) in Computer Software Engineering*

Aug 2024 – May 2026

### Vellore Institute of Technology

Tamil Nadu, India

*Bachelor of Technology (B.Tech) in Information Technology*

Jul 2017 – Jul 2021